

Venkat Reddy Boya | Software Engineer

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PROFESSIONAL SUMMARY

Python Software Engineer with **5+ years of experience** in developing scalable applications, backend systems, and intelligent solutions using **Python, Django, Flask, and RESTful APIs**. Skilled in building data-driven products, integrating SQL/NoSQL databases, and implementing **ML/DL models** with Scikit-learn, TensorFlow, and Keras. Experienced in **automation, ETL pipelines, and cloud deployments** (AWS, Azure) with a focus on reliability and scalability. Strong background in **CI/CD, Agile practices, and TDD**, with proven success in optimizing application performance, reducing downtime, and enabling predictive analytics.

TECHNICAL SKILLS

Languages:	Python, SQL, JavaScript, Shell Scripting
Frameworks & Libraries:	Django, Flask, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, SQLAlchemy, PyTest
Web Technologies:	HTML5, CSS3, JQuery, JSON, XML
Databases:	MySQL, PostgreSQL, Oracle, MongoDB, Redshift
Cloud & DevOps:	AWS (EC2, RDS, S3, Lambda, IAM, CloudWatch), Azure, Terraform, Ansible, Docker, Kubernetes, Jenkins, ELK Stack
Data & ML:	ETL pipelines, Feature Engineering, Supervised & Unsupervised Learning, Deep Learning (ANN, CNN, RNN), NLP, Data Visualization (Matplotlib, Seaborn, Power BI)
Version Control & CI/CD:	Git, GitHub, Bitbucket, Jenkins, Maven
Methodologies:	Agile, Scrum, SDLC, TDD, BDD

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – *Lumen Technologies, USA*

| Feb 2023 – Present

- Designed and deployed end-to-end AI applications using Python, Django, TensorFlow, Keras, and MySQL in Linux, achieving a 25% improvement in deployment speed via Jenkins-based CI/CD pipelines.
- Automated data parsing and validation workflows with Python scripts, improving test coverage by 30% and reducing manual QA time.
- Developed views & templates with Python & Django's view controller and templating language to make a user-friendly website interface.
- Built and consumed RESTful APIs for data exchange, enabling faster integration across systems and cutting API response times by 20%.
- Enhanced data-driven applications by integrating Pandas, NumPy, and MongoDB, improving large dataset handling efficiency by 40%.
- Created application structure based on Angular style guides and best practices.
- Created user interface components using Angular and React.js.
- Implemented ML classification models with Scikit-learn for anomaly detection in system logs, reducing false alerts by 18% and improving monitoring accuracy.
- Created a database using MySQL and wrote several queries to extract data from the database.
- Worked in a NoSQL database on simple queries and writing Stored Procedures for normalization and renormalization.
- Responsible for debugging the project monitored on JIRA (Agile).
- Collaborated with cross-functional teams in agile sprints, delivering scalable features with a 95% average on-time sprint completion rate.
- Performed troubleshooting, fixed, and deployed many bug fixes for applications that were a main source of data for both customers and the internal customer service team.

JR. SOFTWARE ENGINEER – *HCL Tech, India*

| Jan 2019 – Jul 2021

- Developed dynamic AI applications with Python, TensorFlow, Django ORM, and following Layered and Event-driven architecture, improving system modularity and reducing defect rates by 15%.
- Wrote scripts in Python for extracting data from an HTML file.
- Built and optimized ETL pipelines and SQL queries for Redshift, reducing query execution time by 35% and improving reporting efficiency.
- Automated AWS infrastructure tasks (EC2 scaling, RDS monitoring, Lambda triggers) with Python scripts, lowering operational workload by 20%.
- Developed a web-based application using the Django framework with Python concepts.
- Generated Python Django forms to maintain the record of online users.
- Used Django API's to access the database.
- Implemented Machine Learning models (classification & regression) in Python to forecast sales demand, improving the accuracy of business planning by 22%.
- Applied Deep Learning (CNNs with TensorFlow & Keras) for image-based quality checks, improving detection accuracy from 70% to 91% in production use cases.
- Provisioned infrastructure using Terraform and integrated with Jenkins pipelines, achieving consistent deployments across Dev, QA, and Production environments.
- Configured and managed Redshift clusters for data warehousing, optimizing storage and compute costs by 15%.
- Deployed containerized applications with Docker and Kubernetes, reducing deployment errors by 40% and improving scalability.
- Conducted data validation and anomaly detection with SQL and Python, ensuring 99% data accuracy across client reporting systems.
- Implemented SOAP/RESTful web services in JSON format.

- Associated with debugging the applications monitored on JIRA using agile methodology.
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EDUCATION

Master's in Computer Science: *University of Dayton, OH, USA*

| Dec 2022